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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2025-2026

EE2005 – ELECTRICAL DEVICES AND MACHINES

April / May 2026 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Consider the magnetic circuit shown in Figure 1. The mean lengths of the various sections are $l_v = 12$ cm, $l_{hl} = 18$ cm, $l_{hr} = 18.5$ cm, and the cross-sectional areas are $A_v = 4$ cm^2; $A_h = 2$ cm^2, $A_c = 2$ cm^2. The coil wound on the center limb has a resistance of $R = 25\ \Omega$ and consists of $N = 200$ turns. The relative permeability of the left limb, center limb and right limb are $\mu_{rl} = 800$, $\mu_{rc} = 1200$ and $\mu_{rr} = 1200$, respectively. Take the permeability of free space to be $\mu_o = 4\pi \times 10^{-7}$ H/m.

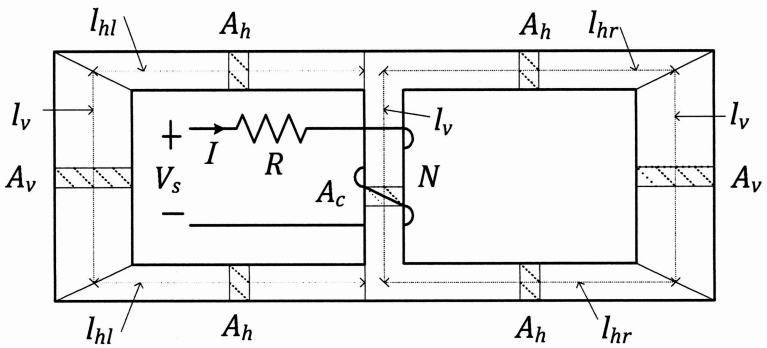


Figure 1

Note: Question No. 1 continues on page 2.

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(i) Draw the magnetic equivalent circuit of the given magnetic structure and show clearly the respective reluctances $\mathcal{R}_l$, $\mathcal{R}_c$, and $\mathcal{R}_r$ for the left limb, centre limb and right limb, respectively. If the flux in the right limb ϕ_r is required to be 72 μWb, determine the respective fluxes ϕ_l and ϕ_c in the left and center limbs. Also, determine the DC current in the coil to establish these fluxes. Ignore any leakage and fringing.

(ii) Determine the respective energies stored in the left, center and right limbs and the inductance of the coil for the operating condition described in part (a)(i).

(iii) If the DC source is now replaced by a 50-Hz AC source, determine the amplitude of the voltage source V_s to achieve twice the flux in the center limb obtained in part (a)(i).

(17 Marks)

(b) Figure 2 shows an electromagnetic system which consists of an immovable part and a movable part both separated by an air gap distance of x m. The force exerted on the movable part is $f = 1$ N. The coil wound on the immovable part has a resistance of $R = 20\ \Omega$ and consists of 300 turns. The cross-sectional areas of the two parts are 1 cm^2 each and the mean lengths of the movable and immovable parts are $l_a = 5$ cm and $l_b = 2.25\pi$ cm, respectively. The relative permeability of the two parts is $\mu_r = 2500$ and the permeability of free space is $\mu_o = 4\pi \times 10^{-7}$ H/m. Draw the magnetic equivalent circuit of the magnetic system and determine the DC voltage source V_s required to maintain the air gap distance at $x = 1$ mm.

(8 Marks)

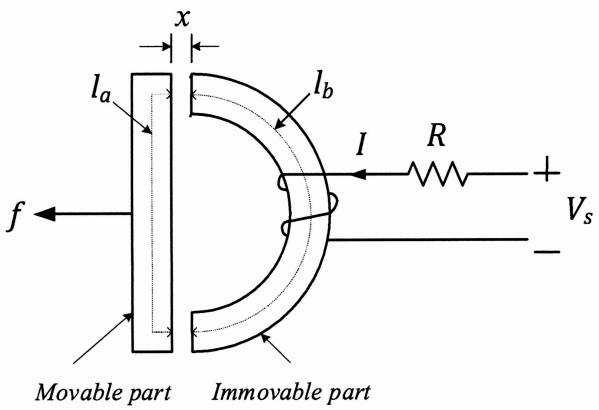


Figure 2

2. Consider a single-phase transformer rated 46 kVA, 2300/230-V and 50-Hz. A short circuit test was conducted with the low-voltage side short circuited. The measurement data obtained are given as follows:

$$V_{sc} = 65\ V, \quad I_{sc} = 20\ A, \quad P_{sc} = 1100\ W$$

Note: Question No. 2 continues on page 3.

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(a) Draw the equivalent circuit of the transformer referred to the low-voltage side and ignore the no-load current. Using this equivalent circuit, calculate the voltage regulation if the transformer is operated at full load with a power factor of 0.85 lagging and at the rated secondary voltage. Take the rated secondary voltage as the reference, i.e., $\vec{V}_2 = 230\angle 0^o$ V. Comment on the performance of the transformer.

(10 Marks)

(b) The high-voltage side of the transformer is now connected to the voltage source V_s through a feeder with a series impedance of $Z_{f1} = 0.25 + j1.2\ \Omega$ and a load of 35 kW at a power factor of 0.85 leading is connected to the low-voltage side of the transformer at 230 V. Draw the equivalent circuit of the transformer referred to the low-voltage side and use it to calculate the sending end voltage and the input complex power at the sending end of the feeder. Take the rated secondary voltage as the reference, i.e., $\vec{V}_2 = 230\angle 0^o$ V.

(7 Marks)

(c) The two-winding transformer is now connected as a step-up autotransformer to give the maximum kVA rating. Draw the diagram of the step-up autotransformer, showing all the relevant voltages and currents. Also, calculate the rated primary and secondary currents of the autotransformer as well as its kVA rating.

(8 Marks)

3. A three-phase induction motor is rated 440 V, 50 Hz, and has 4 poles. It is configured in a Y connection and has the following per-phase parameters referred to the stator.

$$X_1 = 1.1\ \Omega,\ R_2 = 0.6\ \Omega,\ X_2 = 1.6\ \Omega,\ X_M = 100\ \Omega$$

The DC test results are given as follows: 20 V, 18 A. Figure 3 shows the per-phase equivalent circuit of the induction motor referred to the stator.

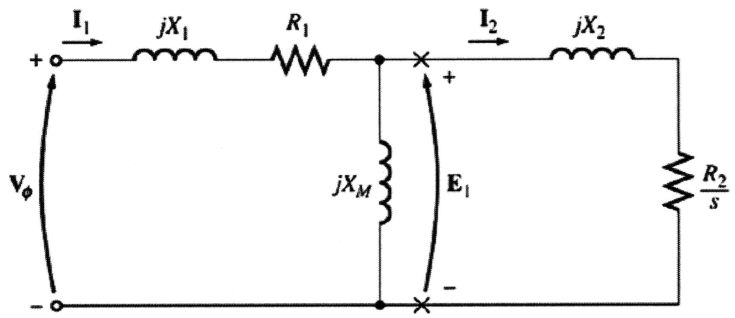


Figure 3

(a) Determine the stator resistance R_1.

(5 Marks)

Note: Question No. 3 continues on page 4.

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(b) Using Thevenin's equivalent circuit, calculate the optimal slip at which maximum torque is developed.

(5 Marks)

(c) Using the optimal slip obtained in part (b), determine the maximum torque developed and the input power factor.

(5 Marks)

(d) Compute the air gap power during motor starting.

(5 Marks)

(e) Assume that X_2 can be adjusted while all other parameters remain unchanged. Determine if there is a possibility of a new value for X_2 that can be used to achieve maximum starting torque.

(5 Marks)

4. A 240-V DC shunt motor has an armature winding resistance of 0.2 Ω and a field winding resistance of 100 Ω, as shown in Figure 4. The rotational loss of the motor is 400 W. At full load, the motor consumes an input power of 12 kW and operates at a speed of 1000 rpm.

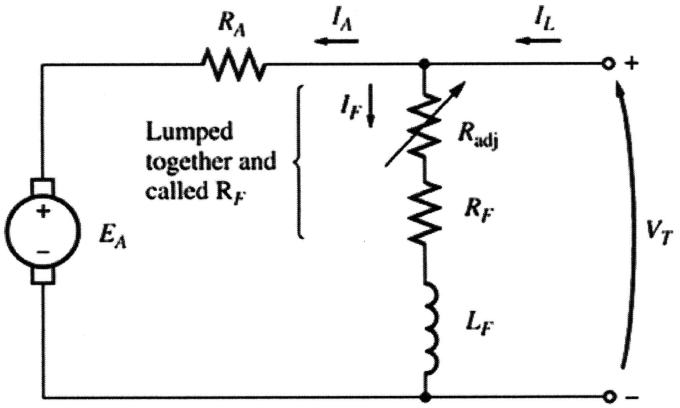


Figure 4

(a) Derive the essential current and voltage equations for the DC shunt motor circuit.

(5 Marks)

(b) Determine the internal generated voltage and the torque developed by the motor at full load.

(5 Marks)

(c) Determine the armature current and torque developed by the motor when it is running at a speed of 980 rpm.

(5 Marks)

Note: Question No. 4 continues on page 5.

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(d) Determine the input current and the output power of the motor if the internal generated voltage is 220 V.

(5 Marks)

(e) If the load is disconnected from the motor shaft, calculate the input current and the internal generated voltage of the motor.

(5 Marks)

END OF PAPER

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EE2005 ELECTRICAL DEVICES & MACHINES

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.